Enterprise Meeting Intelligence Platform

Complete 3-Chat IDE Timeline, Feature Upgrades & Architectural Evolution

Total Development Chats: 3 Sessions	**Core ASR Engine:** Whisper ASR + Bhashini	**Diarization:** 8-Signal / 124-Dimensional
Current Production Version: v3.0	**Pytest Backend Tests:** 23/23 Passed	**Security:** SHA-256 Ledger + GDPR Sanitized

■ CHAT 1: Ingestion Architecture, Latency Acceleration & Real-Time Streaming

1.1 The Challenge & Root Cause:

Audio processing was freezing and taking 3–4 minutes per upload. Investigation revealed a PyTorch Apple Silicon MPS tensor device mismatch (slow_conv2d_forward_mps).

1.2 Upgrades & New Features Implemented in Chat 1:

- **ASR Processing Acceleration (4 min → 1–2 sec):** Switched HuggingFace Whisper ASR execution pipeline to optimized CPU (openai/whisper-tiny / openai/whisper-base), cutting latency to sub-2 seconds.
- **Real-Time 3s WebM Chunk Streaming:** Built /audio/recording/{id}/stream to ingest live WebM audio chunks continuously during live meetings.
- **Synchronized Live Transcript Feed (LiveTranscript.tsx):** Renders spoken dialogue in real-time as attendees speak, appending tokens incrementally.
- **100% Personalized Meeting Minutes Engine:** Overhauled _local_nlp_extract in ai_pipeline.py to extract exact sentence quotes, participant attendee names, exact financial figures ($50,000, 5 Lakhs, INR), and specific resolutions.
- **AI Grounding & Faithfulness Guardrail:** Implemented verify_faithfulness computing grounding_accuracy_score to ensure extracted minutes are backed by transcript tokens.

■ CHAT 2: High-Precision Multi-Signal Diarization Engine (v2.0 → v2.5)

2.1 The Challenge:

Standard diarization failed to accurately detect speaker changes during fast back-and-forth debates, causing speaker turns to merge erroneously.

2.2 Upgrades & New Features Implemented in Chat 2:

- **124-Dimensional Acoustic Feature Vectors:** Combined 20 MFCCs + Delta + Delta-Delta + Pitch (F0) + Spectral Centroid + Zero-Crossing Rate to capture vocal resonance.
- **Sub-Second 0.8s Temporal Windows:** 0.8-second sliding window with 0.2-second stride (10x higher temporal resolution than standard 3s windows).
- **Cosine Metric Agglomerative Clustering:** Implemented adaptive distance thresholding calculated dynamically from audio feature distance statistics.
- **Interactive 1-Click Speaker Renaming:** Clicking speaker labels in SpeakerTimelineView.tsx opens a rename modal that updates all associated turns across the entire meeting transcript globally.
- **12 Controlled Benchmark Scenarios:** Created initial benchmark suite (case_01 to case_12) verifying speaker transitions, pauses, and turn preservation.

■ CHAT 3: Production-Grade Speaker Voice Enrollment & Identity-Aware Diarization (v3.0)

3.1 The Innovation & Core Concept:

Introduced pre-meeting participant voice calibration so the AI learns each person's unique acoustic signature at the start of the meeting and recognizes them accurately during subsequent speaking turns.

3.2 Upgrades & Major New Features Implemented in Chat 3:

- **3-Sample Multi-Criteria Voice Calibration (SpeakerCalibrationView.tsx):** Collects 3 distinct 5–10s audio samples per participant under natural speaking conditions.
- **Interactive Quality Scorecard:** Automated gate checks for SNR (>8 dB), Audio Clipping (<4%), VAD Speech Ratio (>35%), and feature stability with 'Good to Enroll' or 'Please Record Again' guidance.
- **Anti-Spoofing Dynamic Phrases:** Generates randomized dynamic phrases ('The quick brown fox jumps over 42 lazy dogs in Q3') to prevent audio replay attacks.
- **8th Signal Integration:** Embedded Calibrated Identity Match Evidence into the MultiSignalDiarizationEngine alongside the 7 acoustic/VAD signals.
- **4-Tier Identity Decision Matrix:** MATCHED (>=0.72), AMBIGUOUS (0.55-0.72), UNKNOWN (<0.55 - never forces false match), and UNAVAILABLE (declined consent).
- **Biometric Privacy & 1-Click Vector Purging:** Explicit consent toggle, sanitized public DTOs (CalibratedVoiceProfilePublic), and DELETE /api/v1/meetings/{id}/voice-profiles/{id}.
- **Profile Drift Guardrail (>0.35 distance):** Flags acoustic drift as ProfileDriftCandidate for host review rather than silently overwriting enrolled profiles.
- **Human Correction Learning:** When a reviewer corrects an UNKNOWN_SPEAKER tag, the system creates a candidate profile update.
- **'Why This Speaker?' Explanation Modal:** Interactive modal displaying exact evidence (Cosine 94%, Pitch F0 91%, Spectral Centroid 95%, Profile Stability 97%).
- **Enterprise Voice Profiles Dashboard (/voice-profiles):** Dedicated page managing active profiles, quality scores, versions (v1-v3), and drift risk levels.
- **Comprehensive Verification:** 22 benchmark scenarios (case_01 to case_22), 23/23 Pytest backend suites passing, and clean Next.js build.

■ Summary Matrix Across the 3 Project Chats

Milestone Domain	Chat 1 Focus	Chat 2 Focus	Chat 3 Focus (Current v3.0)
ASR / Audio	CPU Whisper fix; 1-2s latency	Sub-second 0.8s chunk alignment	Dynamic phrase anti-spoofing + SNR gate
Live Streaming	3s WebM chunk streaming	Live transcript word alignment	Pre-flight calibration roster check
Diarization	Basic turn splitting	124-dim spectral vectors	8-signal matrix + 4-tier classification
Voice Profiles	None	Single-turn centroid memory	3-sample composite profiles + drift guard
Minutes / NLP	Personalized sentence quotes	Faithfulness grounding score	Decision register + Action items
UI / Studio	Live transcript view	1-click speaker renaming	/voice-profiles + 'Why This Speaker?'
Governance	Basic SQLite records	SHA-256 Audit chain logging	Biometric purging + GDPR sanitization